

12	A	By First law of thermodynamics, besides cooling, the internal energy can also be decreased by mechanical work through expanding the gas, as it does work on its surroundings. Option D is true only for ideal gas, which has zero microscopic PE. Option B is true, except during change of state. E.g. when ice melts, its internal energy is increased because of the increase in its microscopic PE. But there is no change in its microscopic KE, and thus no temperature change. Option C Temperature is a measure of the <u>average</u> (microscopic) KE (not KE+PE).
13	B	By Newton's 2nd law